

Semantic Geometry for Safety-Critical Decisions

From multilingual convergence to safety-relevant semantic constraints

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1. Why This Matters

LLMs already appear in decision-support settings

medical triage, legal assistance, financial analysis, and autonomous agents

Core challenge

We still do not know how exactly these models take decisions, and whether the internal states of these systems preserve the semantic distinctions that safe decisions should depend on, even under different phrasings.

2. Robustness Is Not Enough

Invariant case

Semantically equivalent inputs should imply equivalent decisions.

“Patient with severe penicillin allergy and bacterial infection.”

“Patient allergic to penicillin, presenting with infection.”

Sensitive case

A small but critical semantic modifier should change the decision.

“Patient with infection.”

“Pregnant patient with infection.”

The issue is not generic prompt robustness. The issue is semantic invariance versus semantic sensitivity.

In safety-critical use, hidden-state geometry is part of the reliability problem.

3. Motivation: The Geometry We Need

Desired mapping

- semantically equivalent inputs $\rightarrow$ nearby internal states
- semantically different situations $\rightarrow$ distinct states

Why it matters

- better state abstraction than raw text-conditioned features
- uncertainty and deferral when an input falls outside known semantic regions

$$h_I(x) = s_I(z) + n_I(f) + \epsilon_I$$

We want the semantic term $s_I(z)$ to dominate decisions, while wording, language, and anisotropy remain removable nuisance structure.

It would be useful if we could audit this geometry, and intervene on it if needed

4. Related Work

Saglam et al. 2025

Claim: semantic information lies in low-dimensional linear subspaces.

Method: linear probes and separability tests across 11 autoregressive LLMs.

Schiekiera et al. 2026

Claim: hidden-state geometry aligns with behavioral semantic similarity.

Method: psycholinguistic similarity and free-association experiments, then RSA against hidden states.

Burns et al. 2023

Claim: truth and falsehood appear along approximately linear directions.

Method: linear probing plus activation steering on truthful vs. false statements.

Lee et al. 2025

Claim: embeddings share lower-dimensional manifold structure across models.

Method: intrinsic-dimension estimation and locally linear manifold analysis.

These papers justify a geometric view of meaning, but not yet the safety-critical pair we need: nuisance invariance plus semantic sensitivity.

5. Gap and Research Question

Gap

Most existing studies show that semantic structure exists, but they do not test whether that structure has the properties required for reliable decision-making.

Property A

Invariance to nuisance variation

Property B

Sensitivity to meaningful semantic change

Do language models contain a recoverable semantic space that separates meaning from wording and language, while remaining sensitive to critical semantic modifiers?

6. Methodology: Multilingual Nuisance Analysis

Dataset

- 24 semantic factors across 6 domains
- 5 English surface families, then multilingual expansion
- 5 languages: EN, DE, ES, AR, ZH
- 1248 total sentences in the all-family set
- same factor appears under wording and language changes

Concrete factor example

Same factor, different surface forms:

“The object is hot.”

“Upon contact, one immediately notices the intense heat.”

plus exact-translation variants across the 5 languages.

What we compute

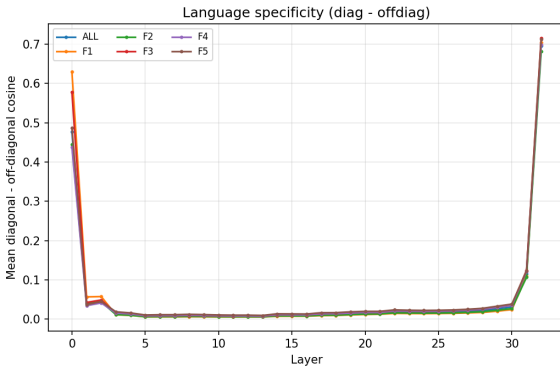
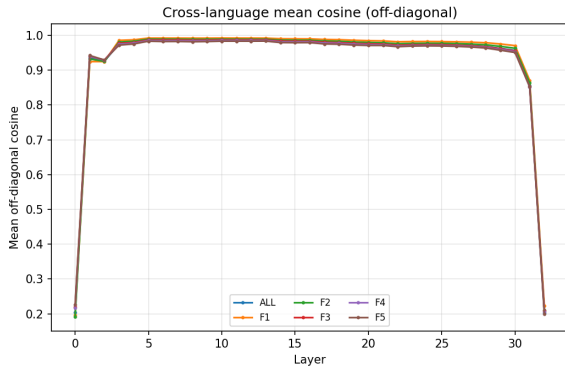
- hidden states at all 33 layers in pythia-2.8B
- pair geometry for exact-translation, same-factor, same-domain, and different-topic pairs
- metrics: cosine similarity and L2 distance

Nuisance control

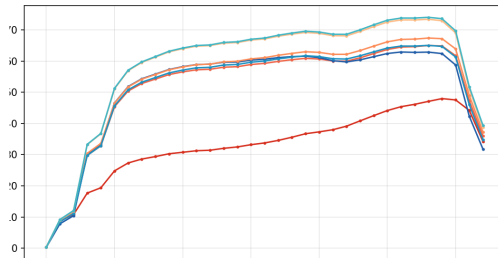
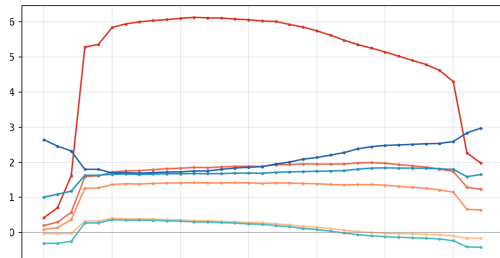
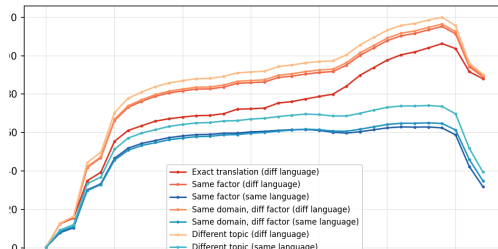
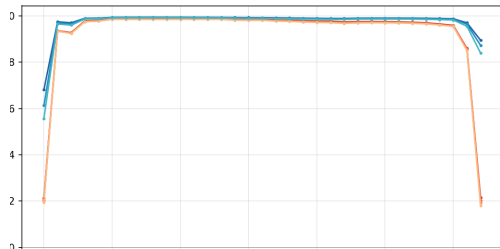
Language residualization is a per-layer linear least-squares removal of a language one-hot regressor. It is not a train/val/test probe. We compare it directly against blind top- k deflation.

7. Same Meaning Converges Across Languages

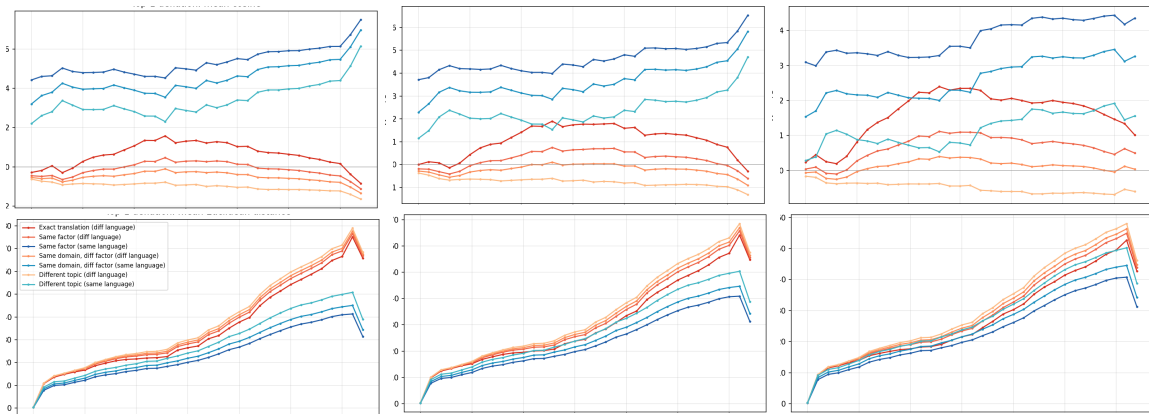
G2. Layerwise cross-lingual convergence indices by family (raw)



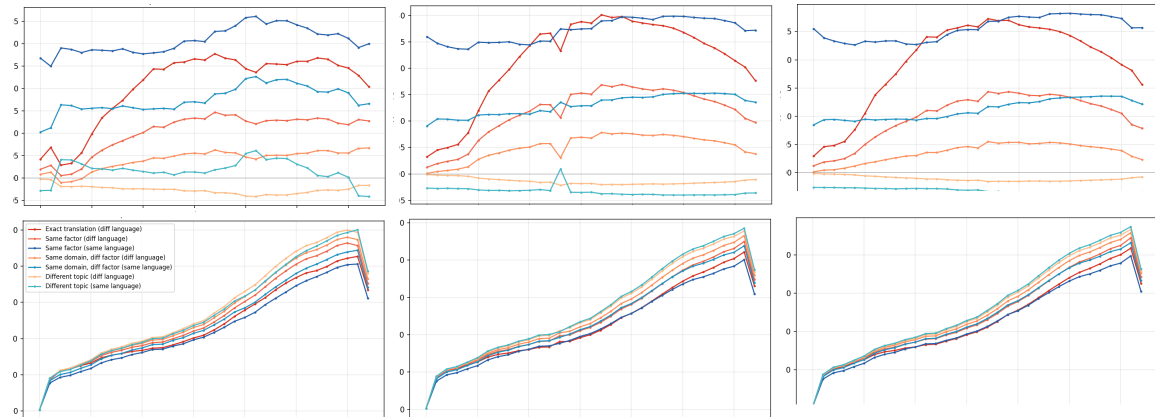
8. Raw Geometry and Language Residualization



9. Blind Deflation: Top-1 to Top-3



10. Blind Deflation: Top-4 to Top-6



11. Methodology: Constraint Sensitivity Analysis

Dataset

- 10 refinement chains and 10 redundant-control chains
- 5 levels per chain, 8 sentences per level
- 800 total sentences
- every level is analyzed separately: no cumulative pooling

Refinement example

form → medical consent form → emergency medical
consent form → pediatric emergency medical consent form
→ notarized pediatric emergency medical consent form

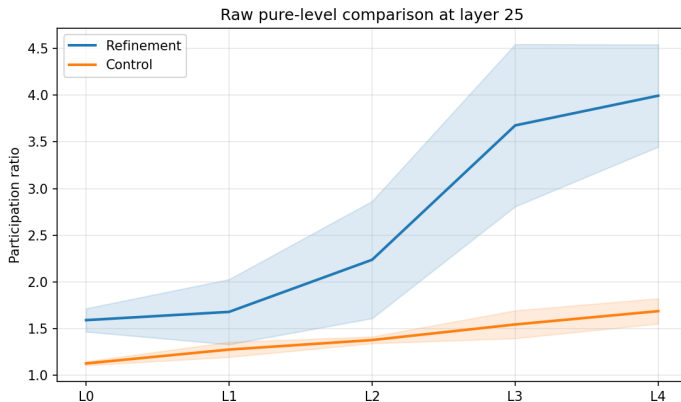
Control example

fire → hot fire → bright hot fire → strong bright hot fire
→ strong bright hot burning fire

What we compute

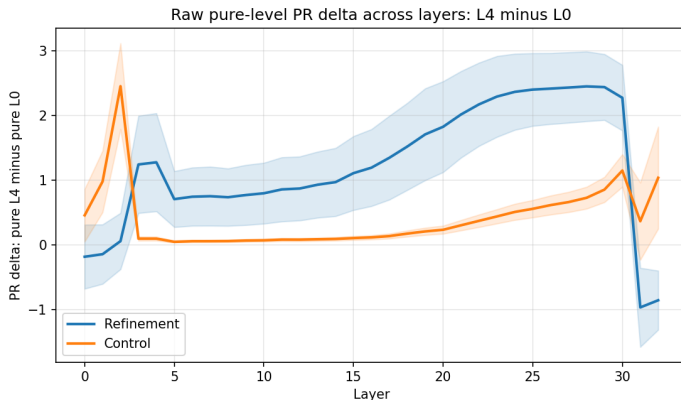
Hidden states are traced across all 33 layers. We compare participation ratio by level, condition, and intervention branch, with raw geometry, language residualization, and top- k dominant-direction deflation.

12. Pure-Level Geometry at Layer 25



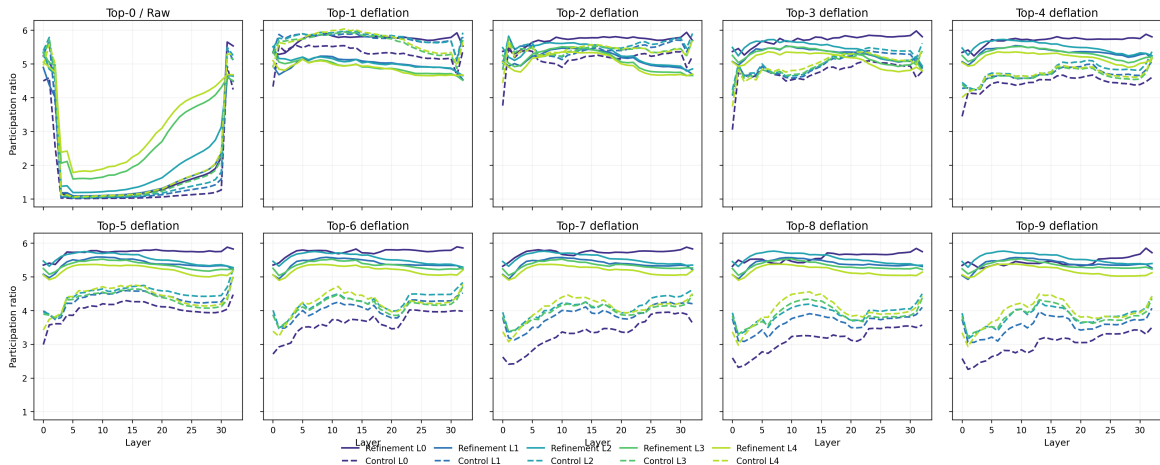
Using only pure levels, the refinement chains rise more steeply than the control chains.

13. The Raw Gap Persists Across Layers

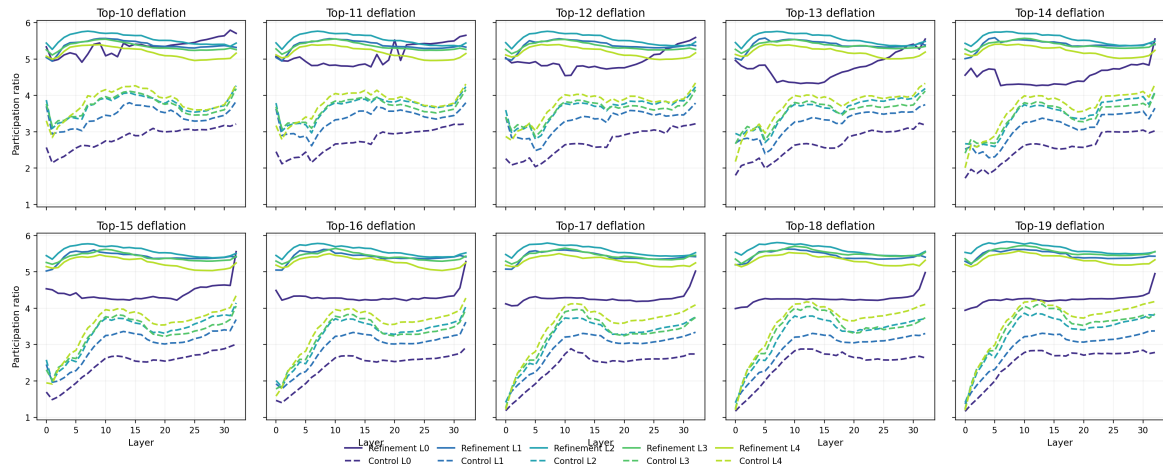


The raw $PR(S4) - PR(S0)$ gap stays positive through much of the network and is strongest in a late-middle layer band.

14. Top-0 to Top-9 Deflation

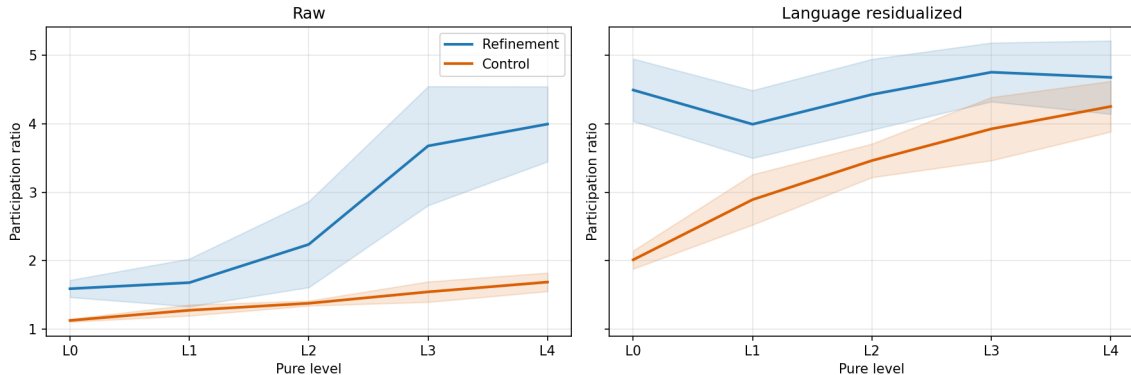


15. Top-10 to Top-19 Deflation



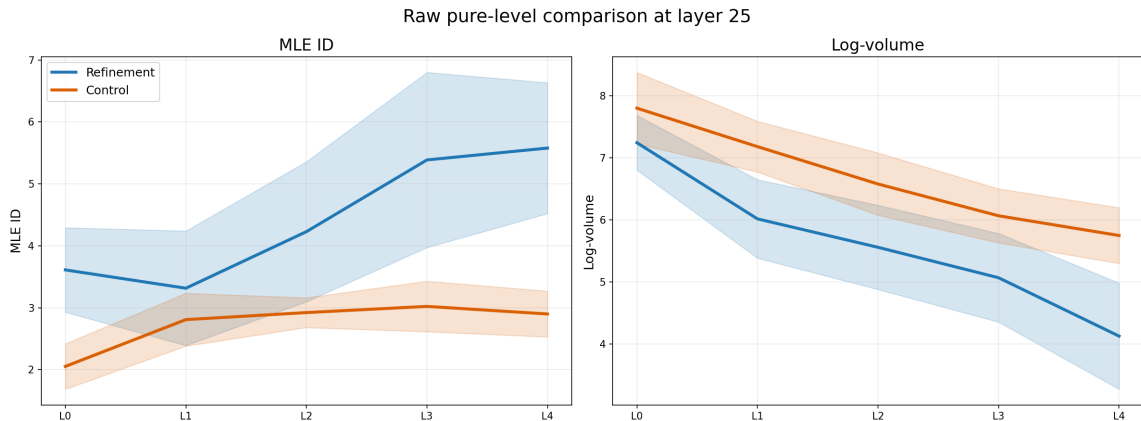
16. Raw vs Language-Residualized Constraint Geometry

Pure-level PR at layer 25: raw and language residualized

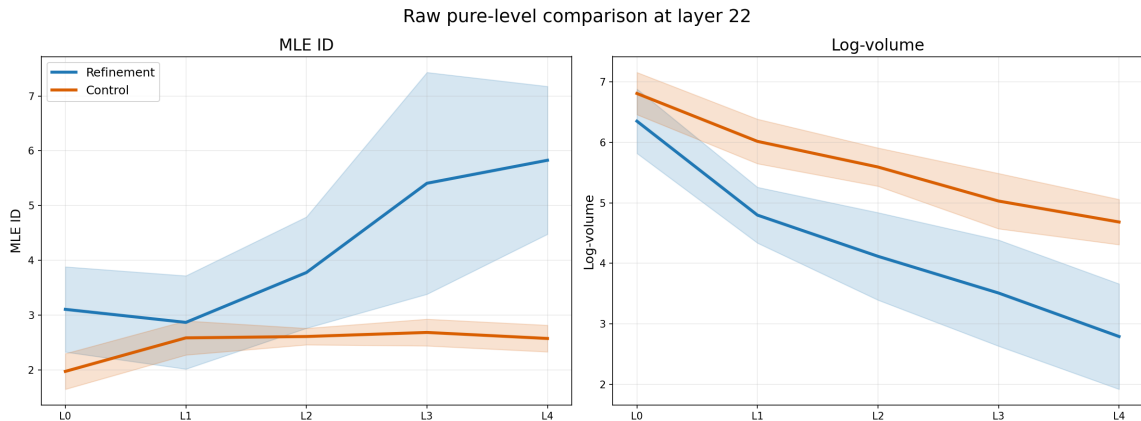


Language residualization is asymmetric: it helps the control chains more at higher levels, but helps the refinement chains mainly at lower levels. Late refinement partly overlaps with language-linked directions.

17. MLE ID and Log-Volume at Layer 25

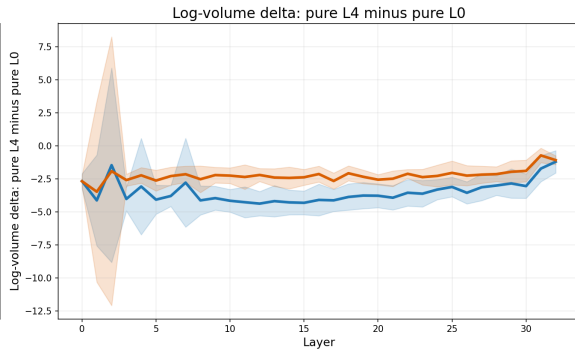
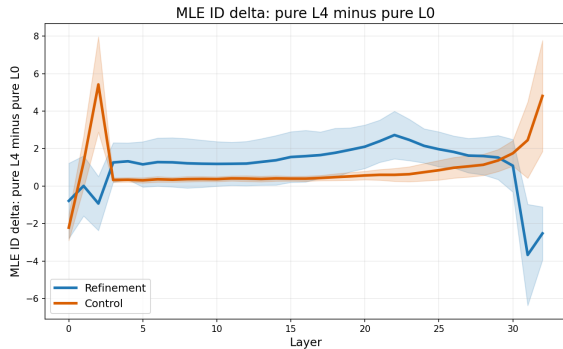


18. MLE ID and Log-Volume at Layer 22

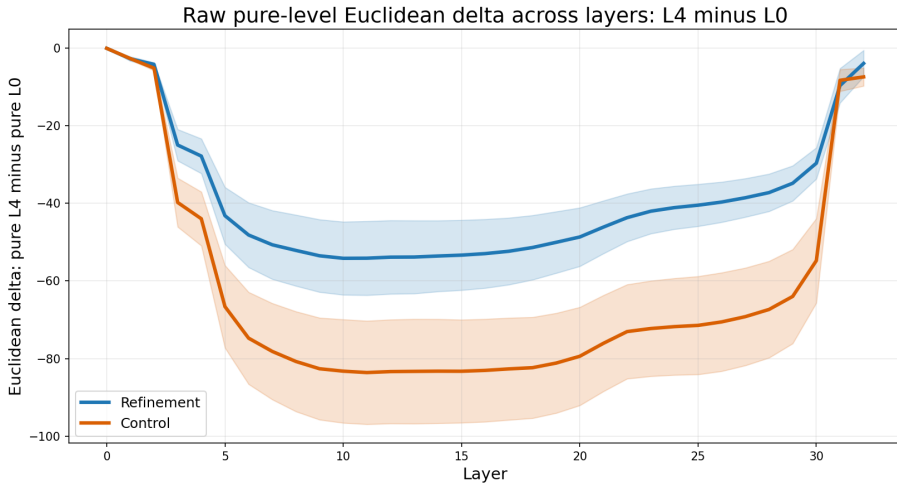


19. MLE ID and Log-Volume Deltas

Raw pure-level delta across layers: L4 minus L0

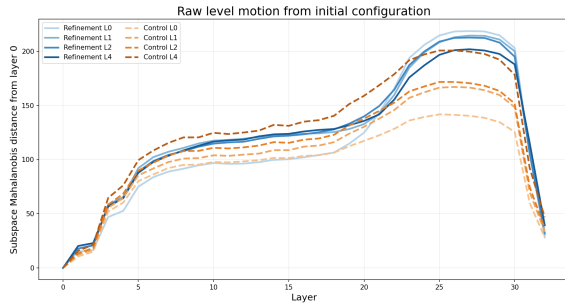
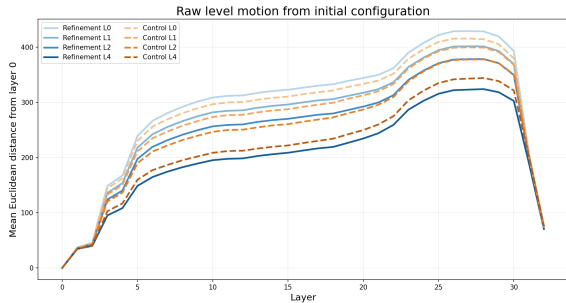


20. Euclidean Delta Across Layers

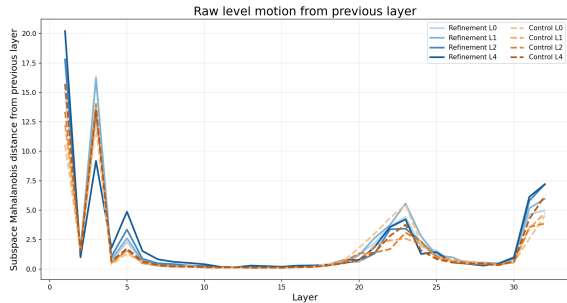
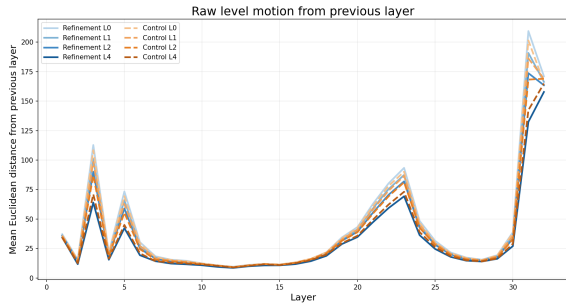


Negative values mean the level-4 cloud is more compact than level 0 in raw Euclidean spread.

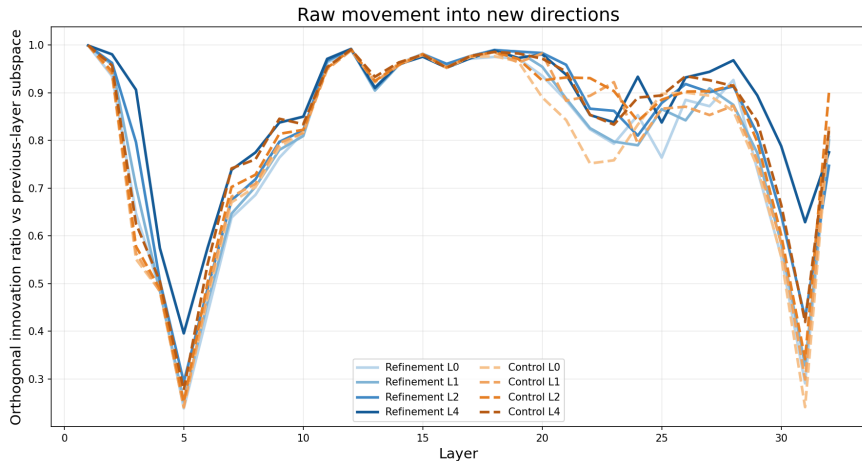
21. Level Motion from Initial Configuration



22. Level Motion from Previous Layer



23. Movement into New Directions



Innovation ratio: $\|\delta_{\perp}\|/\|\delta\|$, with $\delta = h_l - h_{l-1}$ and $\delta_{\perp}$ orthogonal to the previous-layer level subspace.

24. Geometric Conclusions

Semantic base

Stable meaning could be recoverable in a structured low-rank subspace.

Nuisance fibers

Wording, language, and anisotropy act like structured transverse directions, not just random noise.

Constraint spine

Strong refinement moves concepts along a concentrated shared axis rather than uniformly expanding their rank.

The strongest picture is a semantic base manifold with nuisance fibers attached to it, plus a concentrated refinement spine for safety-relevant constraints.

25. Safety Implications and Extensions

Why this matters for safety

- invariance and sensitivity can be audited in representation space before full behavioral deployment
- semantic geometry can support state abstraction, out-of-support detection, uncertainty, and human deferral
- failures of nuisance control are potential failure modes for decision consistency

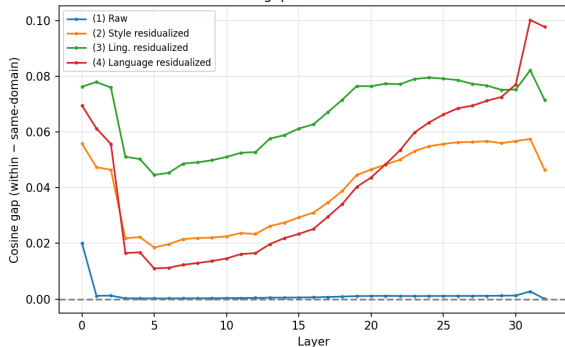
Next steps

local manifold analysis, activation interventions, trajectory-based geometry during generation, and evaluation on richer safety-specific datasets.

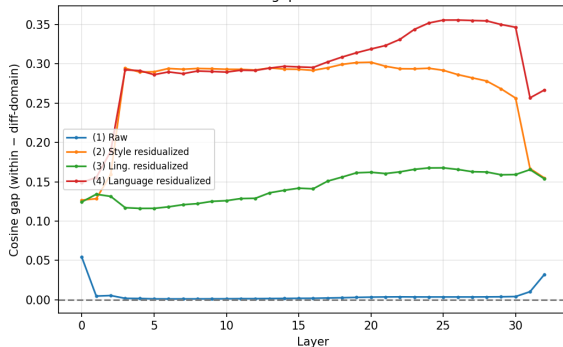
Appendix: Additional Multilingual Geometry View

E. Cosine similarity gaps — 4 residualization conditions

E.1 — Cosine gap: within vs same-domain



E.2 — Cosine gap: within vs diff-domain



Appendix: Language-Dominant Direction Alignment

